

开源软件基础

Git 与版本控制

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Outline

1 Before Git

2 Git 入门

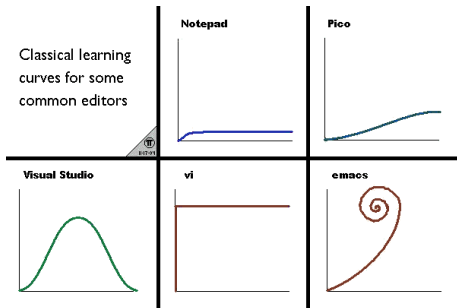
Vim

Why Vim

- 用于提交 commit 信息（非必要）
- 用于解决冲突（非必要）
- 那为什么？

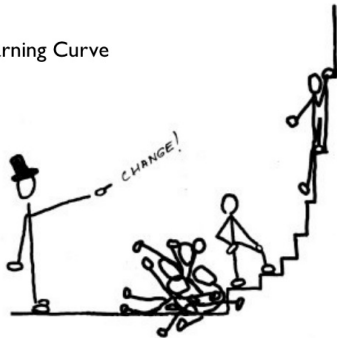
编辑器效率对比

Classical learning curves for some common editors

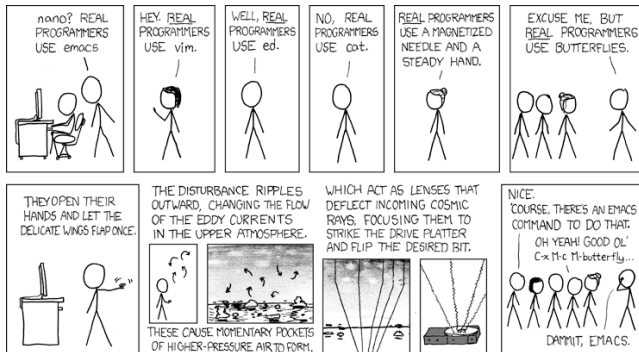


编辑器效率对比

Learning Curve



Real Programmer¹



¹<https://www.xkcd.com/378/>

今天就能开始用的小知识

开源软件中的 Diss 与 Respect

- vi by Bill Joy
- vim by Bram Moolenaar
- neovim by Thiago de Arruda^a

^a[https:](https://geoff.greer.fm/2015/01/15/why-neovim-is-better-than-vim/)

[//geoff.greer.fm/2015/01/15/why-neovim-is-better-than-vim/](https://geoff.greer.fm/2015/01/15/why-neovim-is-better-than-vim/)

Bram Moolenaar on neovim

It's going to be an awful lot of work, with the result that not all systems will be supported, new bugs introduced and what's the gain for the end user exactly?

Total refactoring is not a solution. It's much better to improve what we have. Perhaps with some small refactorings specifically aimed at making Vim work better for users.^a

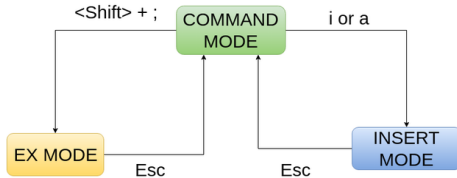
^a[https:](https://groups.google.com/forum/m/#!topic/vim_dev/x0BF9Y0Uby8)

[//groups.google.com/forum/m/#!topic/vim_dev/x0BF9Y0Uby8](https://groups.google.com/forum/m/#!topic/vim_dev/x0BF9Y0Uby8)

高效编辑

- 基于模式
- 高效移动
- 命令组合
- 自定义/绑定

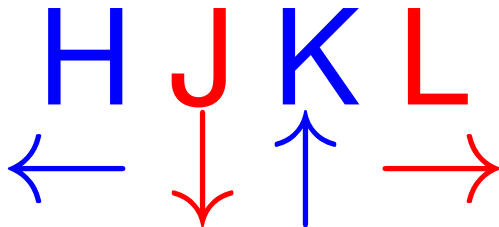
Vim 模式



高效移动



高效移动



高效编辑



移动

- gg: goto first line
- G: goto last line
- 0: goto first column
- ^: goto first non-blank column
- \$: goto last column
- w: next word
- W: next Word
- e: end of word
- E: end of Word
- fa: find a in current line
- Fa: find a in current line backward
- ta: find till before a
- Ta: find till before a backward

单独使用命令

- i: insert
- I: insert at front
- a: append
- A: append after end
- o: append new line
- O: insert new line

命令与文本对象

命令

- c: Change
- d: Delete
- y: Yank
- gU: Upper case
- gu: Lower case
- g~: Switch case ...

文本对象

- aw: a word
- iw: inner word
- ab: a bracket
- ib: inner bracket

合体（误）

命令 + 移动/文本对象

- **dj**: delete current and the next lines
- **y0**: copy to the line head
- **ciw**: change inner word
- **dab**: delete a bracket
- **gUit**: change inner tag to lower case ...

今天就能开始用的小知识

表妹

CNLabelContactRelationYoungerCousinMothersSiblingsDaughterOrFathersSistersDaughter^{ab}

▲ euske 5 months ago [-]

Well, there's CNLabelContactRelationYoungerCousinMothersSiblingsDaughterOrFathersSistersDaughter, which is two letters longer than CNLabelContactRelationElderCousinMothersSiblingsDaughterOrFathersSistersDaughter.

▲ yorwba 5 months ago [-]

表姐, in case you were wondering.

▲ zeroimpl 5 months ago [-]

Looks like this constant should be parsed as: ElderCousin && (MothersSiblingsDaughter || FathersSistersDaughter)

▲ ttflee 5 months ago [-]

Mother's Sibling's Daughter or Father's Sister's Daughter.

FYI, Mother's Sibling is 舅舅 (Mother's Brothers) or 姨 (Mother's Sisters), and Father's Sister is 姑姑.

▲ perlwle 5 months ago [-]

二舅 Mother's second Brother?

^a <https://developer.apple.com/documentation/contacts/cnlabelcontactrelationyoungercousinmotherssiblingsdaughterorfatherssistersdaughter>

^b <https://news.ycombinator.com/item?id=20341855>

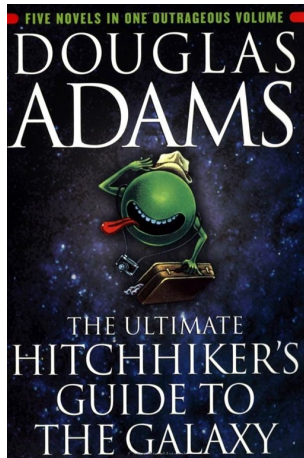
命令行模式

相对高阶功能

- `:%s/re.pattern/replace/g` 全局替换
- `:g/re.pattern/d` 删除匹配行
- `:h! E478 Don't panic`
- `:h 42`

What is the meaning of life, the universe and everything? *42*
Douglas Adams, the only person who knew what this question really was about is now dead, unfortunately. So now you might wonder what the meaning of death is...

书籍推荐（伪）



Practise makes perfect

继续学习

- vimtutor
- vimcasts^a
- 7 Habits For Effective Text Editing^b
- Maybe emacs^c

^a<http://vimcasts.org/>

^bhttp://v.youku.com/v_show/id_XMTIwNDY5MjY4.html?spm=a2h0k.8191407.0.0&from=s1.8-1-1.2

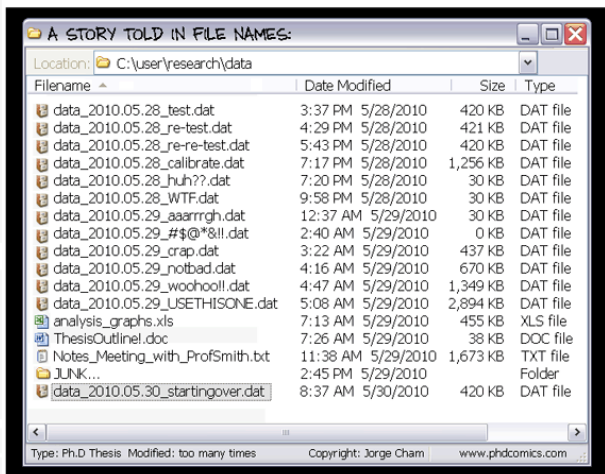
^c<https://www.bilibili.com/video/av6048518/?from=search&seid=12030402565005031700>

Outline

1 Before Git

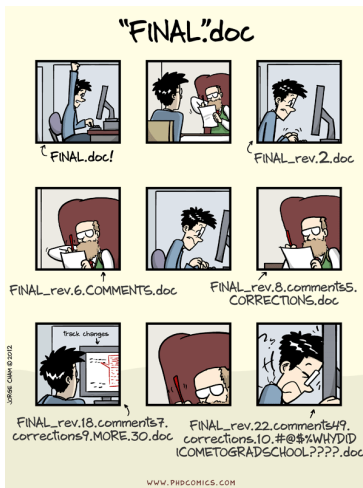
2 Git 入门

Why Git²



²<http://phdcomics.com/comics/archive.php?comicid=1323>

Why Git³



³<http://phdcomics.com/comics/archive.php?comicid=1531>

Git

git 是一个分布式版本控制软件，最初由林纳斯·托瓦兹创作，于 2005 年以 GPL 发布。最初目的是为更好地管理 Linux 内核开发而设计。git 最初的开发动力来自于 BitKeeper 和 Monotone。很多著名的软件都使用 git 进行版本控制，其中包括 Linux 内核、X.Org 服务器和 OLPC 内核等项目的开发流程。

今天就能开始用的小知识

Git 名字的由来

林纳斯·托瓦兹自嘲地取了这个名字“git”，该词源自英国俚语，意思大约是“混账”。^a

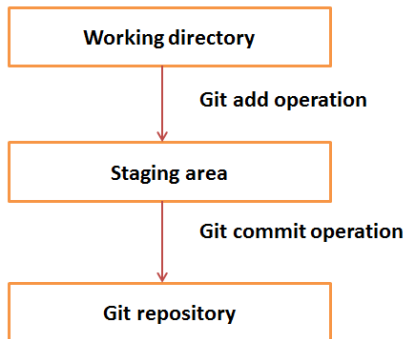
I'm an egotistical bastard, and I name all my projects after myself. First Linux, now git.

^a<https://git.wiki.kernel.org/index.php/GitFaq>

Git 命令初步



Git 命令初步



Git 命令初步

```
#initialization
$ git init

#configuration
$ git config --global user.name "Your Name"
$ git config --global user.email "you@your.server"

# First commit
$ git add example.py

# adds file to the staging area
$ git commit -m "First commit"

# Second commit
$ git add another.py

# adds file to the staging area
$ git commit -m "Second commit"
```

Branching and Merging

- By default you are on the "master" branch
- To create a new branch and switch to it:
 - `$ git branch branchName`
 - `$ git checkout branchName`
- Merge changes back to master branch
 - `$ git checkout master`
 - `$ git merge branchName`
 - To delete branch
 - `$ git branch -d branchName`

My Typical Git Workflow

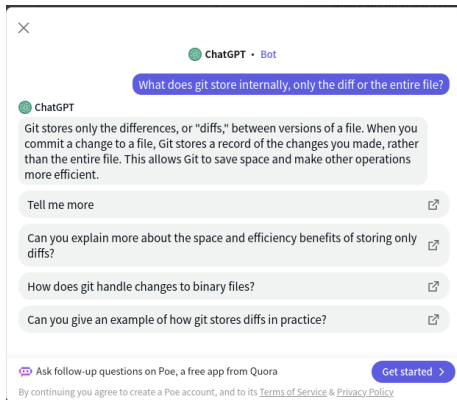
- Master branch is always clean and working
- When I want to try a new idea, create a branch
- Work and commit to that branch until code is clean and working
- Switch back to master and merge branch into master
- Delete branch

Things to Keep in Mind

- Any branch can be merged to any other branch
 - Careful, this might get confusing
- Must commit before switching branches
- Unless you add it, a file will not be tracked.
- If you want to ignore a file, add it to the `.git/info/exclude` file.
- To revert a file back to it's state at the last commit:
 - `git checkout fileName`
- To revert the entire repo to it's state at the last commit:
 - `git reset --hard HEAD`
 - CAREFULL! You'll loose all your changes since your last commit!
- <http://git-scm.com/>

Things to Keep in Mind

- git 存储整个文件，而非 diff
- 网络搜索并不可靠



How git Works

```
$ git init
已初始化空的 Git 仓库于 /tmp/demo/.git/
$ echo hello world > readme
$ git add readme
$ find .git/objects/
...
.git/objects/3b
.git/objects/3b/18e512dba79e4c8300dd08aeb37f8e728b8dad
$ git cat-file -p 3b18e5
hello world
$ git commit -m init
[master (根提交) 4722e8d] init
 1 file changed, 1 insertion(+)
 create mode 100644 readme
$ git cat-file -p 4722
tree 2c5264370f2630da80ee38f412213d59657af6e8
author Zhilei Ren <zren@dlut.edu.cn> 1788934139 +0800
committer Zhilei Ren <zren@dlut.edu.cn> 1788934139 +0800

init
$ git cat-file -p 2c52
100644 blob 3b18e512dba79e4c8300dd08aeb37f8e728b8dad      readme
$ git cat-file -t 2c52
tree
```

How git Works

```
$ echo name > contributors
$ git add contributors
$ git commit -m contrib
[master 73ca5d0] contrib
 1 file changed, 1 insertion(+)
 create mode 100644 contributors
$ git cat-file -p 73ca
tree b0ebaa6f7d727e9ad7dd9853307339ef584da826
parent 4722e8df1fbbf644cc5a89944c157adf8c902e30
author Zhilei Ren <zren@dlut.edu.cn> 1788934283 +0800
committer Zhilei Ren <zren@dlut.edu.cn> 1788934283 +0800

contrib
$ git cat-file -p b0eb
100644 blob f121bdbbff4df6ff03e927c8d84e230da55fb1c0b    contributors
100644 blob 3b18e512dba79e4c8300dd08aeb37f8e728b8dad    readme
```